

PersistQA: Benchmarking Identity Grounded Question Answering in Long Videos

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Abstract

Following one person through a long video and answering questions about what they wear, do, and touch is a basic requirement for video understanding. Existing long video benchmarks do not test it, because they reward coarse temporal gist. We introduce PersistQA, a benchmark of 3,667 eight-option multiple-choice questions over 449 long videos. Every question is grounded in a referent identity that must be tracked across shots, scenes, and appearance changes. We evaluate 22 models across open-weight full-fidelity VLMs, proprietary API models, and long video models that compress visual tokens. The benchmark is far from solved. The best model scores 28.3%, no question is answered by every model, and 22.9% are answered by none. Three findings overturn the natural reading of these numbers. First, difficulty is not identity confusion. Per-video accuracy is uncorrelated with the number of people on screen and with identity-switch opportunity, and 20 tracker plus re-identification pipelines that do the identity bookkeeping never beat feeding the raw video. Second, temporal coverage is the dominant lever, not raw pixel fidelity. Accuracy climbs monotonically with frame count for full-fidelity models, yet token-compression models stay flat near 19% no matter how many frames they read. Lowering per-frame resolution from 448 to 112 pixels barely changes accuracy, so compression models fail by discarding query-relevant content rather than by lowering fidelity. Third, temporal tracking skills sit at chance. Role swaps score 13.9% and multi-hop tracking 16.1%, while single-scene appearance questions are easiest. Model scale barely helps, with a parameter-to-accuracy correlation of 0.05. PersistQA isolates a capability that neither scale nor current long video architectures deliver.

1 Introduction

Re-identification, recognizing the same person or entity across time and viewpoints, underpins applications in video surveillance, forensic search, sports analytics, and person-following robotics (Ye et al., 2022; Zheng et al., 2015; Davila et al., 2023). It is studied as a dedicated retrieval problem with strong embedding models (Zhou et al., 2019; He et al., 2021; Li et al., 2023b; Chen et al., 2023) and increasingly under hard real-world shifts such as clothing change (Liang & Rawat, 2025a; Yang et al., 2021; Qian et al., 2020; Wan et al., 2020) and video (Zheng et al., 2016; Yu et al., 2024; Kim et al., 2025). Yet these systems operate on cropped galleries, while real video understanding requires re-identification fused with question answering inside untrimmed footage. This paper asks whether end-to-end video models carry that fused skill.

Long video understanding is usually framed as a context-length problem, where fitting more frames into the model is expected to raise performance. Recent long video models therefore compress visual tokens so that hundreds of frames fit in one context (Song et al., 2023; He et al., 2024; Li et al., 2024c; Shen et al., 2024; Shu et al., 2024; Liu et al., 2025; Li et al., 2025). Popular long video benchmarks reward this design, since their questions ask about plot, scene, or global content (Fu et al., 2024; Zhou et al., 2024; Wang et al., 2024b; Wu et al., 2024), and strong systems now report above 85% on them (Liu et al., 2025; Li et al.,

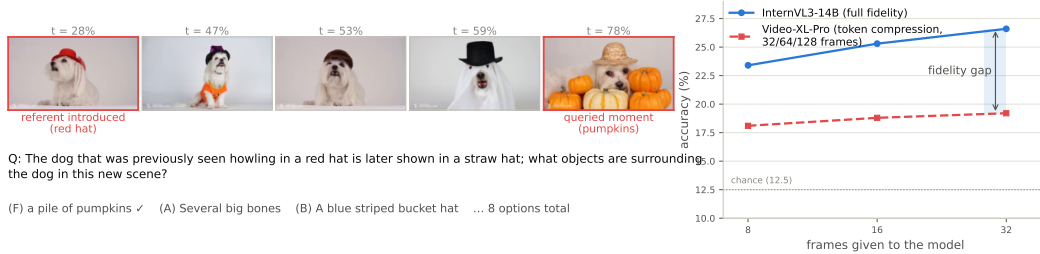


Figure 1: The PersistQA task and the headline finding. Left, a real benchmark item names a referent at one moment and asks about that referent after a scene cut, which forces re-identification followed by fine-grained reading. Right, a full-fidelity model (InternVL3-14B) improves with frame count while a token-compression model (Video-XL-Pro) stays near 19%, a fidelity gap above 7 points.

2025). These numbers hide a basic skill. A model may summarize a video yet fail to lock onto one person, follow that identity across cuts, and read what the person does.

We call this skill identity-grounded question answering. It is the dominant question form in real long video analysis. A user asks what the man in the blue jacket does after he leaves the podium, or which object the woman who entered last picks up. Answering requires re-identification, which means finding the same identity again after cuts, occlusions, and appearance changes (Zheng et al., 2015; Zhou et al., 2019; Davila et al., 2023), together with fine-grained perception at the right instant. Neither skill alone is sufficient.

PersistQA contains 3,667 eight-option multiple-choice questions over 449 long videos. Every question names a referent by appearance, location, or activity, then asks about an attribute of that referent at another point in the video. We tag each question along four axes. These are the perceptual capability probed, an assigned difficulty, the referral strategy, and a ten-way taxonomy of the re-identification challenge that ranges from single-shot reference to cross-scene re-identification, multi-hop tracking, appearance change, occlusion recovery, and role swaps. Multiple-choice options leak language priors (Goyal et al., 2017; Agrawal et al., 2018; Buch et al., 2022), so a committee of thirteen models answers every question without the video, and questions the committee beats are removed or repaired. Text-only accuracy after filtering is 12.6%, which matches the 12.5% chance level, so measured performance reflects vision.

We evaluate 22 models. These span fifteen open-weight full-fidelity VLMs across six families (Zhu et al., 2025; Bai et al., 2025; Gemma Team, 2025; Lu et al., 2024; Lin et al., 2023; Li et al., 2025), three proprietary API models, and four long video token-compression models (Shen et al., 2024; He et al., 2024; Shu et al., 2024; Liu et al., 2025). The strongest model reaches 28.3%, roughly double chance and far from solved. No question is answered by all 22 models, and 22.9% are answered by none.

Our contribution is a diagnosis, not only a leaderboard. A benchmark named after re-identification invites the reading that models fail by confusing identities. We test that reading and reject it. Per-video accuracy is uncorrelated with the number of people on screen and with identity-switch opportunity, and twenty pipelines that ground the referent with dedicated re-identification models never beat the raw video. The bottleneck is coverage, not raw fidelity. More frames help every full-fidelity model, token-compression models stay flat near 19% across a sixteen-fold frame increase, and lowering per-frame resolution to 112 pixels barely moves accuracy. The ten-way taxonomy then localizes the failure, with role swaps at 13.9% and multi-hop tracking at 16.1% against easiest types near 20.6%. Scale does not close the gap, since the parameter-to-accuracy correlation is 0.05. We release the benchmark, every model prediction, the shortcut audits, and the analysis code. Section 5 sketches a re-identification-aware method that follows directly from the diagnosis and keeps the benchmark as the core contribution.

2 Related Work

Long video models and token reduction. Video LLMs began by aligning a vision encoder to a language model on short clips (Lin et al., 2023; Li et al., 2023a; Maaz et al., 2024; Li et al., 2024a; Wang et al., 2024a; Bai et al., 2025; Chen et al., 2024c; Zhu et al., 2025; Yao et al., 2024; Ye et al., 2024). Scaling to long video forces a token budget that recent systems meet in different ways. MovieChat carries a dense-to-sparse memory across frames (Song et al., 2023); MA-LMM augments the model with a long-term memory bank (He et al., 2024); LongVU compresses spatiotemporally with adaptive frame selection (Shen et al., 2024); Video-XL and Video-XL-Pro reduce tokens by summarization and reconstructive compression (Shu et al., 2024; Liu et al., 2025); VideoChat-Flash applies hierarchical compression (Li et al., 2025); LLaMA-VID uses two tokens per frame (Li et al., 2024c); and Koala conditions on key frames (Tan et al., 2024). HierarQ modulates features with a task-aware hierarchical querying transformer at a fixed token rate per frame (Azad et al., 2025), and query-agnostic pruning methods allocate tokens by content (Choudhury et al., 2025; Bolya et al., 2023; Chen et al., 2024a; Shang et al., 2024). We show why uniform compression fails on identity-grounded reading, and Section 5 builds on query-conditioned selection (Zhang et al., 2025; Choudhury et al., 2025).

Long video benchmarks. Existing suites evaluate global comprehension through plot, ordering, and coarse event questions (Fu et al., 2024; Zhou et al., 2024; Wang et al., 2024b; Wu et al., 2024; Mangalam et al., 2023; Li et al., 2024b; Yu et al., 2019; Patraucean et al., 2023). Reported accuracies of strong systems now exceed 80% on several of them. A model family that reports 87.5% on original MovieChat-1k questions scores below 20% on PersistQA built from the same source videos. Needle-in-a-haystack protocols test retrieval of a planted item, which measures coverage rather than fine-grained identity reading. PersistQA differs by conditioning every question on tracking one referent across time.

Language priors in multiple choice. Multiple-choice video QA is vulnerable to text-only shortcuts, and language-only models often score far above chance (Goyal et al., 2017; Agrawal et al., 2018; Gururangan et al., 2018; Buch et al., 2022; Chen et al., 2024b). Single-frame baselines further solve many video questions without temporal reasoning (Buch et al., 2022). We remove these shortcuts through committee filtering and report the full text-only and single-frame batteries with the benchmark.

Person re-identification. Re-identification is a mature retrieval field surveyed by Ye et al. (2022). TransReID introduces a pure transformer re-identification backbone (He et al., 2021); SOLIDER learns human-centric representations with semantic supervision (Chen et al., 2023); CLIP-ReID adapts vision-language pretraining to re-identification (Li et al., 2023b); OSNet learns omni-scale identity features (Zhou et al., 2019); and Market-1501 and MEVID supply the image and video galleries these models use (Zheng et al., 2015; Davila et al., 2023). Recent work targets hard shifts. DIFFER disentangles identity from clothing via semantic cues (Liang & Rawat, 2025a); DisenQ disentangles biometric and non-biometric cues with a Q-Former (Liang & Rawat, 2025b); cloth-changing benchmarks LTCC, PRCC, and VC-Clothes stress appearance invariance (Qian et al., 2020; Yang et al., 2021; Wan et al., 2020); and video benchmarks and methods such as MARS, TF-CLIP, and KeyRe-ID target temporal aggregation (Zheng et al., 2016; Yu et al., 2024; Kim et al., 2025). Temporal association is handled by multi-object trackers (Bewley et al., 2016; Zhang et al., 2022; Aharon et al., 2022; Maggolino et al., 2023; Du et al., 2023). We ask a different question, namely whether end-to-end VLMs possess this skill inside natural long video QA, and whether attaching dedicated re-identification pipelines helps. It does not, as Section 4.2 shows.

3 The PersistQA Benchmark

Source and referents. PersistQA is built over 449 long videos from the MovieChat-1k test collection, with a median length of 3.2 minutes and a median of 53 shot cuts per video. Movie footage cuts rapidly, so cross-shot identity persistence is native to the domain rather

Table 1: Composition of PersistQA along its capability and re-identification challenge axes. The full distributions of all four axes appear in Appendix A.

Capability		Re-identification challenge	
Location	849	Cross-scene re-identification	1092
Activity	750	Single-shot reference	515
Appearance	721	Multi-hop tracking	504
Interaction	696	Appearance change	442
Object state	651	Long-term tracking	384
		Action sequence	328
		Spatial relationship	265
		Disambiguation, similar	100
		Occlusion recovery	26
		Role or position swap	11

than an added perturbation. Referents are people in about three quarters of questions and animals in the rest.

Questions and tags. Each question has eight options and one correct answer, with gold letters balanced across positions. We tag every question along four axes, summarized in Table 1. The capability axis covers appearance, activity, location, interaction, and object state. The difficulty axis has 702 easy, 1362 medium, and 1603 hard questions, calibrated against measured accuracy in Appendix D. The referral axis records how the referent is named. The challenge axis is a ten-way taxonomy of the re-identification skill required, which separates single-shot reference, cross-scene re-identification, long-term tracking, multi-hop tracking, appearance change, action sequences, spatial relationships, disambiguation among similar identities, occlusion recovery, and role or position swaps.

Protocol. Models receive the question, the eight options, and frames sampled uniformly from the full video. The standard budget is eight frames, and we ablate 16, 32, 64, and 128. Models output one option letter, parsed by a shared routine and scored by exact match. Every model runs zero-shot, and no model is trained on PersistQA videos or questions.

4 Main Results and Analysis

Table 2 reports the leaderboard. Three facts frame the analysis.

The benchmark is hard and unsaturated. The strongest model is the proprietary Gemini-3.1-Flash-Lite at 28.3%, and the best open model is InternVL3-14B at 26.6% when given 32 frames. Several widely used models sit within a few points of the 12.5% chance level. On the common subset answered by all 22 models, no question is solved by every model and 22.9% are solved by none, shown in Figure 2a. The distribution of solver counts sits far left of the independent-chance expectation, which indicates a large genuinely unsolved core rather than uniform noise.

Scale does not buy the skill. The Pearson correlation between log parameters and accuracy is 0.05 across all models and 0.16 within full-fidelity open models, shown in Figure 2b and by family in Appendix J. Scaling inside a family adds little, since InternVL3 gains 4.3 points from 2B to 14B, and family dominates, since InternVL3-2B at 19.1% beats three 8B models from other families.

Long video specialists trail. The four token-compression models occupy the bottom of the board between 10.4% and 19.2%, despite reading 128 frames against the default eight. Section 4.4 attributes this to a fidelity cost of compression rather than a data or scale artifact.

Each subsection states a question, runs a test, reports the number, and draws one implication. Correlations are Pearson. We read correlation r below 0.3 as weak, 0.3 to 0.7 as moderate, and above 0.7 as strong. Binned curves drop bins with fewer than 20 items, and intervals are bootstrap percentile intervals over questions.

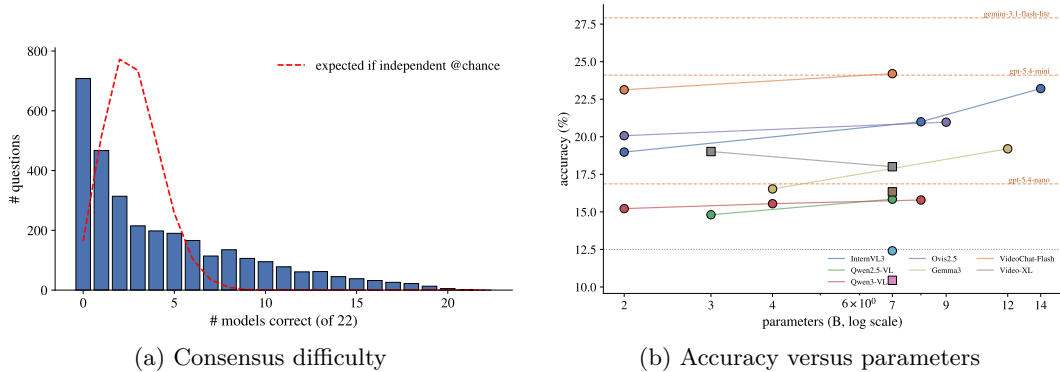


Figure 2: Left, the number of the 22 models that answer each question, against the independent-chance expectation. No question is solved by all models and 22.9% by none. Right, accuracy against parameter count, where scale is a weak predictor at $r = 0.05$ and family dominates.

Table 2: PersistQA leaderboard at each model’s evaluation frame budget. Accuracy is against a 12.5% chance level, best per group in bold. Per-model N varies slightly, because API content filters and undecodable files cause skips, and common-subset comparisons appear in Section 4.

Model	Size	Frames	Acc (%)
Proprietary models			
Gemini-3.1-Flash-Lite	–	8	28.3
GPT-5.4-mini	–	8	24.4
GPT-5.4-nano	–	8	17.0
Open-source VLMs (full-fidelity frames)			
VideoChat-Flash-7B	7B	8	24.2
InternVL3-14B	14B	8	23.4
VideoChat-Flash-2B	2B	8	23.3
InternVL3-8B	8B	8	21.3
Ovis2.5-9B	9B	8	21.1
Ovis2.5-2B	2B	8	20.3
Gemma3-12B	12B	8	19.4
InternVL3-2B	2B	8	19.1
Gemma3-4B	4B	8	16.7
Qwen2.5-VL-7B	7B	8	16.0
Qwen3-VL-8B	8B	8	16.0
Qwen3-VL-4B	4B	8	15.7
Qwen3-VL-2B	2B	8	15.3
Qwen2.5-VL-3B	3B	8	14.9
Video-LLaVA	7B	8	12.4
Long-video models (token compression)			
Video-XL-Pro	3B	128	19.2
Video-XL	7B	128	18.2
LongVU (Qwen2)	7B	1 FPS	16.4
MA-LMM	7B	20	10.4
Chance			12.5

4.1 Which re-identification skills do models lack?

The benchmark is named for re-identification and sits far below ceiling, so the first question is which re-identification skills break. The challenge taxonomy exposes a stable skill hierarchy across all 22 models (full per-model heatmap in Appendix, see Figure 12). Role and position swaps score 13.9% against a 12.5% chance level, and multi-hop tracking scores 16.1%. Long-term tracking and disambiguation among similar identities follow at 18.1%. Single-scene and appearance types are easiest near 20.6%. Models can recognize a described person, since single-shot and appearance questions are the most answerable. Models cannot maintain identity through temporal structure, since chains of references, swaps, and long gaps are hardest. These are the cases where the answer depends on frames that uniform sampling

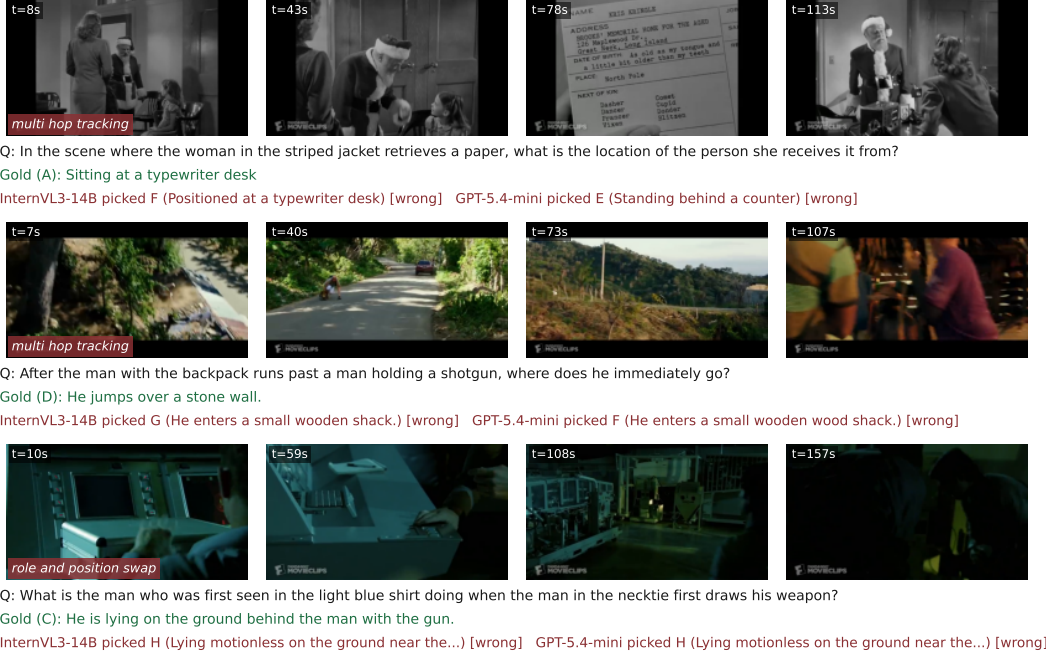


Figure 3: Three failures on tracking-heavy questions. Frames appear left to right, with the question, gold answer, and the letters chosen by InternVL3-14B and GPT-5.4-mini below each row.

tends to miss. Figure 3 shows three failures, all missed by InternVL3-14B and Gemini-3.1-Flash-Lite.

4.2 Difficulty is not identity confusion

The hardest skills in Section 4.1 all involve tracking, which points to identity confusion as the cause. We test that hypothesis three ways and reject each. More people on screen do not make questions harder. Per-video accuracy averaged over models correlates at $r = 0.02$ with concurrent-person count and at $r = 0.02$ with an identity-switch-opportunity score, shown across every measured video property in Appendix G (Figure 17). The only moderate correlates are duration and visual variability, both at $r = 0.42$ and positive, so longer and more varied videos are slightly easier. Difficulty on PersistQA is not explained by opportunities for identity confusion. Question wording, referral, and referent type do not concentrate difficulty either (Appendix H), nor is it tied to any visual-content cluster (Appendix M).

Doing the identity bookkeeping for the model does not help either. We build a three-stage pipeline that detects and tracks every person and animal with four trackers, links tracklets across shots into global identities with generic CLIP features or one of four dedicated re-identification models, grounds the referent to one identity, and renders a clip that marks only that identity. Each of the twenty pipelines runs with all seventeen local models, reported in Appendix E (Table 3). Every pipeline sits at or below the 19.7% raw-video baseline. Tracker choice moves accuracy by at most 0.6 points, and dedicated re-identification models are slightly worse than generic CLIP linking. The strongest VLMs lose up to 6 points, because cropping to one identity discards scene context they exploit. Identity-tracking quality is not the limiting factor.

Errors are systematic. When models err, 52% of wrong answers land on a single modal distractor, against 14% expected under uniform error, and the pairwise correlation between model correctness vectors averages $\phi = 0.24$ with a maximum of 0.54, shown in Appendix D (Figure 14b). Models share a failure mode rather than making independent identity mix-ups.

4.3 Mechanistic probes

Four probes on InternVL3-8B, a representative open full-fidelity model, explain why coverage governs accuracy and raw resolution does not. We instrument the model with attention hooks and read the attention from the generated answer tokens back to the visual tokens of each frame, averaged over heads. Figure 4a shows that this attention concentrates on the last frames and places only 3 to 5% of its total mass on visual tokens, so the model both misses relevant moments through sparse sampling and underweights the frames it does receive. The recency bias holds at 8 and 32 frames and persists even when the final frame is a closing card rather than content, which is why simply adding frames does not repair accuracy. This matters for the coverage story because it shows that unread frames, not unavailable frames, cap accuracy.

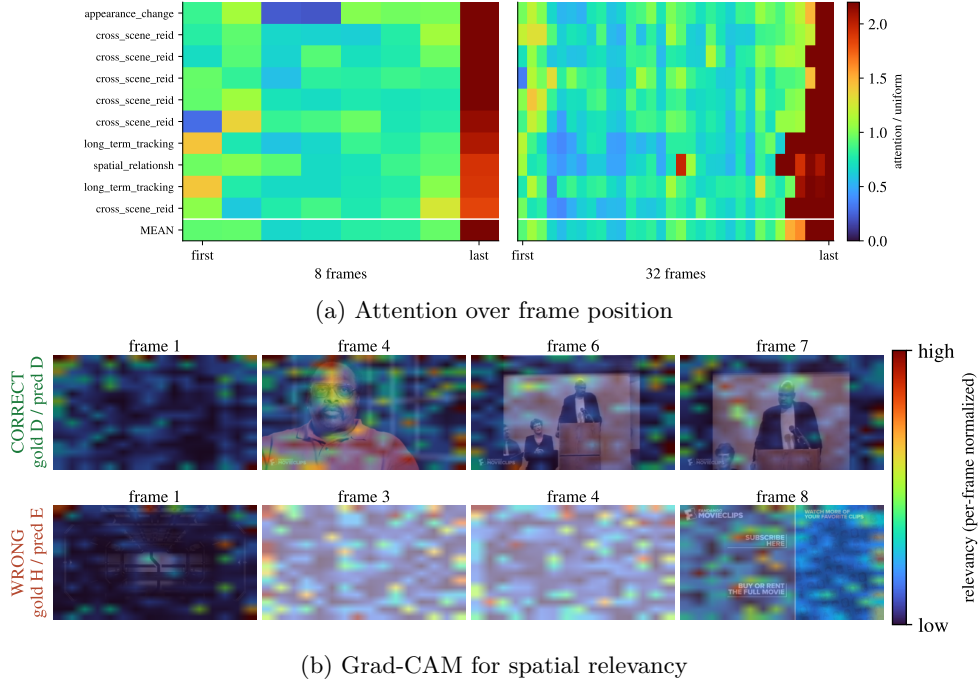


Figure 4: (a) Per-question attention from answer tokens to each frame at 8 and 32 frames, sorted by last-frame mass with the mean row at the bottom. Attention concentrates on the final frame regardless of content. (b) Spatial relevancy on the peakiest correct and wrong examples, per-frame normalized. Correct answers place hot spots on the referent, wrong answers scatter to background and outro cards.

We then map attention onto the native sixteen by sixteen vision patch grid to see where inside each frame the model looks. Figure 4b overlays this relevancy on the peakiest correct and wrong examples. Correct answers place hot spots on the referent, while wrong answers scatter across background and outro cards, which is the spatial counterpart of the frame-level recency bias. Figure 15 embeds a 64-frame pool per video with a CLIP encoder and marks the clusters an 8-frame and a 32-frame uniform sample reach, which visualizes the coverage gap that Figure 6c quantifies and explains why more frames recover missed scene content. Figure 5 tracks the cosine similarity between visual-token representations at reduced budgets and the 32-frame reference across layers, and the dip in middle layers shows where missing coverage propagates into the representation. Together the four probes trace one causal chain, namely sparse sampling misses scene clusters, recency-biased attention underreads the frames that arrive, spatial relevancy lands off the referent, and the representation gap opens in the middle layers. All four run on one representative model, so they explain a mechanism rather than establish a population law, and the chain is read as an existence proof that unread frames, not unavailable frames, cap accuracy. Attention mass is read from generated answer tokens back to visual tokens and averaged over heads, which understates any single sharp head but is robust to head-level noise.

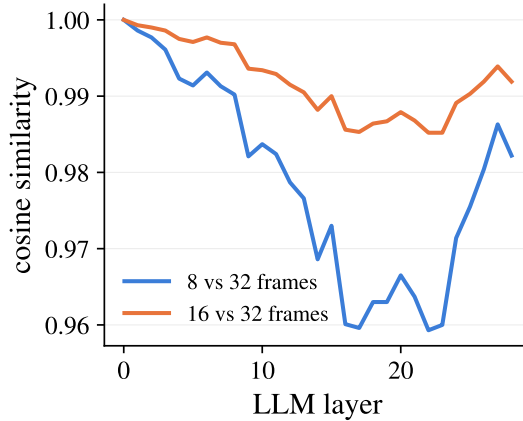


Figure 5: Cosine similarity of visual-token representations at reduced frame budgets against the 32-frame reference, layer by layer. The dip in middle layers marks where missing coverage propagates into the representation.

4.4 The bottleneck is coverage, not raw fidelity

If identity confusion is not the cause, the alternative is that models never see or never resolve the right pixels. This subsection isolates the two halves of that alternative, temporal coverage and per-frame fidelity. Coverage helps monotonically. Raising uniform frames from 8 to 16 to 32 lifts every full-fidelity model tested, and InternVL3-14B goes 23.4, 25.3, 26.6, shown in Figure 6a and across seven models in Appendix I. Compression buys coverage by destroying the fidelity this task needs. Video-XL-Pro reads 32 to 512 frames through reconstructive compression, yet its accuracy stays between 18.1% and 19.2% across the sweep, so a sixteen-fold frame increase adds nothing. Thirty-two full-fidelity frames beat 512 compressed frames by more than 7 points.

Raw per-frame resolution is a weak lever. Holding the budget at 32 frames and lowering resolution keeps accuracy flat from 448 to 112 pixels, and it drops up to 2.8 points only at 56 pixels for the larger models, while InternVL3-2B stays flat from 15.8% to 15.6%, shown in Figure 6b. Compression models therefore do not fail from low pixel resolution. They fail by spending the token budget on the wrong content, which the coverage and selection results establish. Two mechanisms explain the coverage loss. Uniform sampling covers only 57.3% of a video’s scene clusters at 8 frames, 75.3% at 16, and 89.2% at 32, shown in Figure 6c and confirmed by the t-SNE clusters in Figure 15.

Selecting the right frames beats sampling more of them. A training-free probe scores 64 candidate frames against the question with CLIP and keeps the top eight, which adds 1.0 to 2.7 points on three families at a fixed eight-frame budget. Selection compounds with coverage, since selected 16 and 32 frames beat uniform 16 and 32, and the best configuration reaches 29.4%, shown in Figure 6a. The same external selection hurts the one model with native internal frame handling, since VideoChat-Flash drops 3.0 points, so selection must live inside the model. An oracle over five frame strategies reaches about 40%, which bounds how much of the remaining error is a frame-choice problem.

4.5 Necessity and robustness

The controlled probes below run on four models that expose frame-level control, namely InternVL3-2B, InternVL3-8B, InternVL3-14B, and Gemma3-12B. Qwen2.5-VL-7B and VideoChat-Flash-7B are excluded because they handle frame sampling internally and returned no valid predictions under frame manipulation.

Coverage matters only if the questions genuinely need temporal evidence, so we test necessity first and robustness second. The task requires temporal evidence. Moving from 1 to 8 frames lifts InternVL3-2B from 11.0% to 17.2%, a gain of 6.2 points, and InternVL3-8B from 11.8% to 16.0%, a gain of 4.2 points, shown across challenge types in Figure 7. A single frame scores near the 12.5% chance level, so PersistQA answers the single-frame-bias critique of

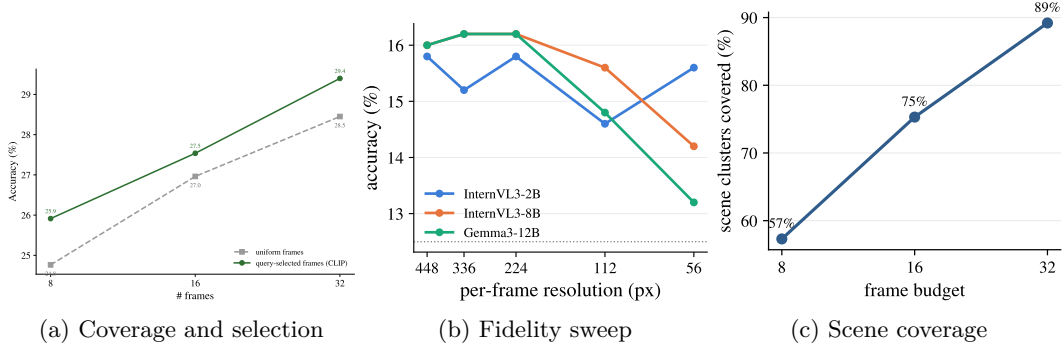


Figure 6: Coverage decides performance, raw resolution does not. Left, query-selected frames beat uniform frames at every budget on InternVL3-14B. Middle, accuracy is flat as per-frame resolution drops from 448 to 112 pixels at a fixed 32-frame budget. Right, uniform sampling covers a shrinking fraction of a video’s scene clusters as the budget falls.

video QA (Buch et al., 2022). Coverage helps unevenly across challenge types, so added frames pay off most where the answer is spread across time and least where a single reference frame already carries it, quantified per challenge type in Appendix K.

Models use temporal order, but weakly. Shuffling the eight frames lowers InternVL3-2B by 1.8 points and reversing them lowers it by 2.8 points, shown in Figure 8a. The effect shrinks with model size and reaches about zero for InternVL3-14B. Order therefore carries a weak signal that larger models rely on less, which matches the tracking failures in Section 4.1.

Answers are fragile to input degradation and option order. Blur, JPEG, and Gaussian noise lower accuracy by up to 4.5 points, moving InternVL3-2B from 17.8% clean to 13.2% under noise, while downscaling to 112 pixels is nearly lossless, shown per model in Figure 8c. Rotating the option contents by three positions leaves accuracy near flat, yet the model changes its content answer on 49.8% of questions, shown in Figure 8b, so the choice is unstable even when accuracy looks stable.

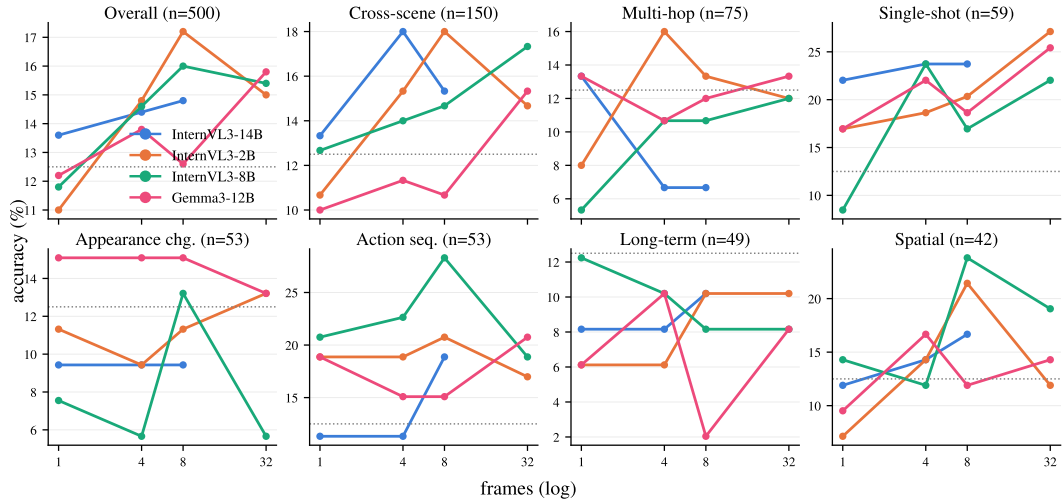


Figure 7: Temporal necessity by challenge type, a small-multiples grid over the four frame-control models. Each panel plots accuracy against the frame budget for one challenge type with its question budget. Every panel rises from 1 frame, so a single frame is insufficient, and the size of the gain varies by challenge type.

These three probes appear in full in Figure 8, with per-model breakdowns and interpretation in Appendix L.

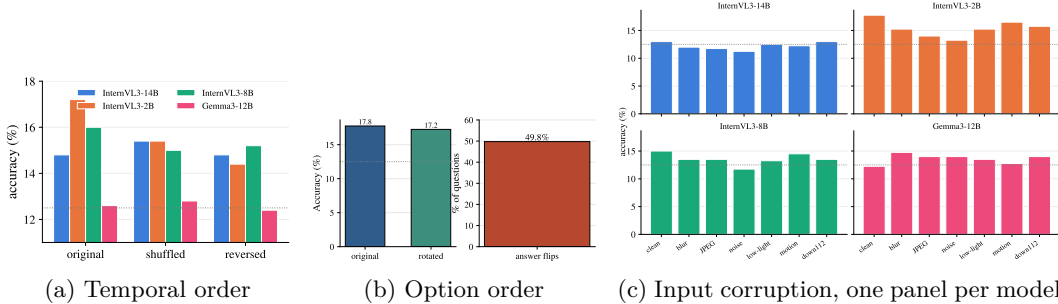


Figure 8: Robustness batteries on the four frame-control models. (a) Shuffling or reversing the eight frames lowers accuracy by 1.8 to 2.8 points on InternVL3-2B and by about zero on InternVL3-14B. (b) Rotating the option contents by three positions leaves accuracy near flat while changing the chosen content answer on 49.8% of questions. (c) Six corruptions at a fixed reduced-resolution probe setting, so panels report relative drops, lower accuracy by up to 4.5 points while downscaling to 112 pixels is nearly lossless.

4.6 Benchmark quality

The findings above are only trustworthy if the benchmark itself is well-behaved, so we close with three checks on stability, significance, and cost. Rankings are stable across evaluation budgets. The Spearman correlation between leaderboards at 8, 16, and 32 frames is 0.95 to 0.99, so conclusions do not depend on the frame budget. Headline numbers carry bootstrap intervals. Gemini-3.1-Flash-Lite at 28.3% has a 95% interval of 26.9 to 29.9, above the interval of GPT-5.4-mini at 22.9 to 25.9, so its lead is significant. Role and position swaps have an interval of 4.1 to 25.2 that includes chance, which confirms that slice sits at chance. Evaluating a proprietary model on the full benchmark costs between 0.5 and 7 US dollars, so PersistQA is practical for repeated use.

5 Toward Closing the Gap

The diagnosis implies a mechanism. Coverage requires compression and compression destroys the fidelity this task needs, so the token budget should be spent non-uniformly and conditioned on the referent and question. We test this as an inference-time change to Video-XL-Pro, keeping its 128-token budget but allocating dense frames around moments a CLIP score marks as query-relevant. On a matched 463-question subset this raises accuracy from 14.0% to 16.0%, a 2.0-point gain with no training. The gain motivates a learned importance head that keeps referent tokens dense while compressing the rest, trained on 16,808 held-out videos with zero benchmark overlap. We report the training-free result here and leave the learned method to future work, so the benchmark remains the contribution.

6 Conclusion

PersistQA measures a skill that long video systems are assumed to have and do not, namely following one identity through a long video and reading fine-grained attributes about it. The benchmark is debiased against language priors, richly tagged, and unsaturated. The analysis rejects the identity-confusion reading, since dedicated re-identification pipelines do not help and identity-confusion proxies do not predict difficulty. The limit is the joint allocation of temporal coverage and per-frame fidelity that current architectures trade against each other, and temporal tracking skills define the frontier new methods must attack.

Reproducibility statement

Every evaluation runs zero-shot with a shared prompt, parser, and sampler. We release per-model predictions for every question, the analysis code that produces every figure and table, and the shortcut audits. The videos come from a public collection and are referenced by identifier. The method of Section 5 is trained only on a disjoint video set with a released overlap audit.

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A Extended composition

PersistQA is designed to spread questions across perceptual skills and re-identification challenges rather than concentrate on any single type. Table 1 lists the capability and challenge axes, and the referral axis records how each question names its referent, with 1001 mixed-reference, 906 appearance-reference, 901 activity-reference, 477 location-reference, and 375 environment-reference questions. Referents are people in 74% of questions and animals in 26%, which keeps the benchmark from collapsing to face recognition. The difficulty axis is assigned at construction time and later audited against measured accuracy in Appendix D.

Figure 9 shows the video and question distributions. The median video runs 3.2 minutes with 53 shot cuts, the median maximum concurrent-person count is 2, and the median option length is 7 words. The high cut rate is the reason cross-shot identity persistence is intrinsic to the domain, since a referent introduced in one shot is typically re-shown after several intervening cuts. Option lengths are tightly distributed around 7 words, which limits the length cue that a model could exploit, and the residual gold-is-longest rate of 16.8% reported in the main text reflects this control. Every distribution is computed over the full released set of 3,667 questions and 449 videos. The challenge counts are deliberately uneven, since the axis mirrors how often each re-identification event occurs in movie footage rather than an imposed uniform prior. Rare types such as role or position swap at 11 questions and occlusion recovery at 26 questions are reported as their own slices rather than merged into neighbors, which keeps a hard slice from being diluted, at the cost of wider bootstrap intervals on those slices.

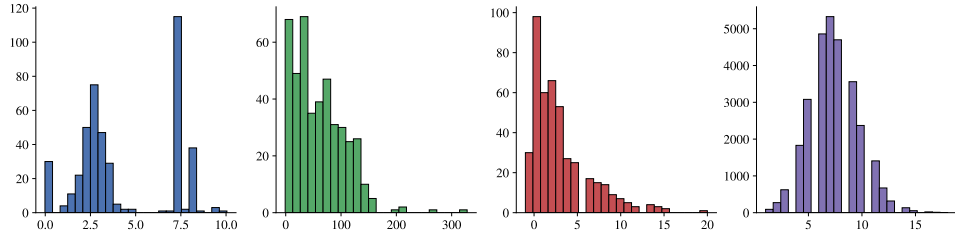


Figure 9: Video and question statistics, namely duration, shot cuts, concurrent people, and option length.

B Metadata axis definitions

Table 1 names the capability and re-identification challenge axes, and this section defines every value so a reader can reproduce the tagging. The capability axis records which perceptual attribute of the referent a question probes.

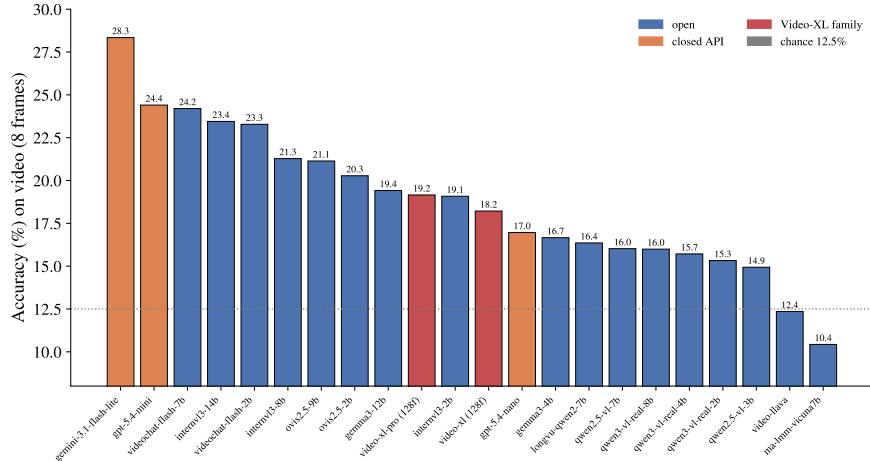


Figure 10: PersistQA leaderboard as grouped bars, colored by model family. Chance is 12.5%.

- **Appearance.** What the referent looks like, including clothing, color, accessories, and physical traits.
- **Activity.** The action the referent performs, such as walking, sitting, throwing, or speaking.
- **Location.** Where the referent is, including the room, setting, or place within the scene.
- **Interaction.** How the referent engages another person or object, such as handing, holding, or facing.
- **Object state.** The state or attribute of an object the referent uses or stands near, such as open, lit, or empty.

The re-identification challenge axis records the kind of identity tracking a question requires, ordered from the least to the most temporal structure.

- **Single-shot reference.** The referent is introduced and asked about inside one continuous shot, so re-identification is minimal.
- **Cross-scene re-identification.** The referent is introduced in one scene and asked about after a cut to a different scene.
- **Long-term tracking.** The referent must be followed across a long temporal gap between introduction and query.
- **Multi-hop tracking.** The referent is reached through a chain of references, for example the person who spoke to the man in red.
- **Appearance change.** The referent changes clothing or accessories between introduction and query, so appearance alone is unreliable.
- **Action sequence.** The answer depends on an ordered sequence of actions the referent performs over time.
- **Spatial relationship.** The answer depends on the spatial relation between the referent and another person or object.
- **Disambiguation, similar.** Several visually similar identities are present and the referent must be told apart from its lookalikes.
- **Occlusion recovery.** The referent is occluded or leaves the frame and must be re-found when it reappears.
- **Role or position swap.** Two identities exchange roles or screen positions and must be kept distinct through the swap.

Every question carries exactly one capability tag and one challenge tag, assigned from the dense per-video annotation and verified in the same re-tagging pass that produced the frozen release.

C Leaderboard in chart form

Figure 10 presents the leaderboard of Table 2 as grouped horizontal bars, which makes the three-way split visible at a glance. Gemini-3.1-Flash-Lite leads at 28.3%, the full-fidelity open models occupy a broad middle band, and the token-compression models cluster at the bottom despite reading far more frames. The bars also show how tightly the field packs between 15% and 24%, which is the reason the benchmark discriminates methods rather than saturating. Each bar uses the same per-model frame budget as Table 2, so bar length is directly comparable to the tabulated accuracy. Coloring by family makes the within-family spread visible and shows that no single family owns the middle band.

D Difficulty calibration and per-slice accuracy

We compute every per-slice number as the mean accuracy over all 22 evaluated models, which smooths idiosyncrasies of any single model and reflects the difficulty the benchmark poses to the field. Assigned difficulty labels track measured accuracy weakly and non-monotonically, at 20.6% easy, 17.5% medium, and 19.2% hard, shown in Figure 13. The

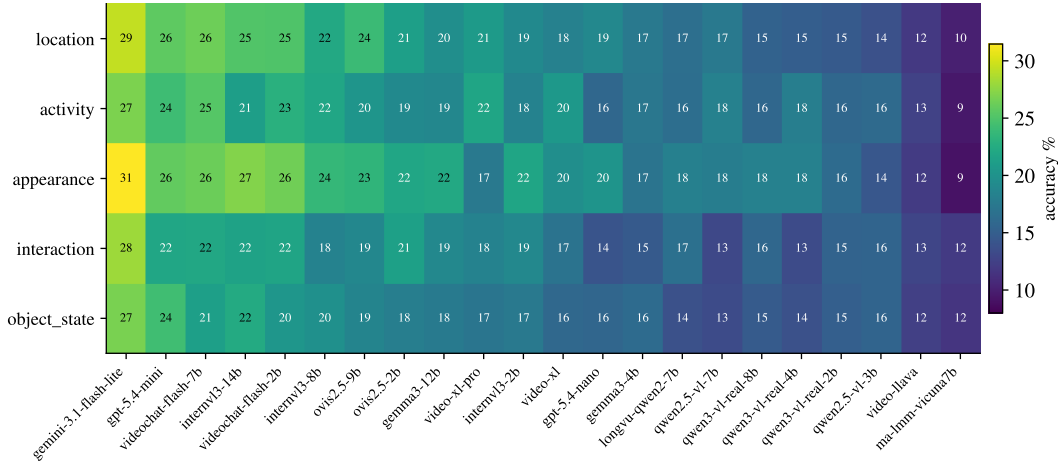


Figure 11: Accuracy of each model by probed capability. The five capabilities span a narrow band, so difficulty is not driven by any single capability but by the re-identification demand cutting across all of them.

non-monotonicity means the labels are useful for coarse stratification but not for fine ranking, so every difficulty-conditioned analysis in the main text uses measured consensus difficulty, defined as the number of models that answer a question correctly, rather than the assigned label.

Figure 11 breaks accuracy down by probed capability. The five capabilities span a narrow band, which shows that difficulty is not driven by one capability but by the re-identification demand that cuts across all of them. Figure 14a shows the predicted-letter distribution per model against the balanced gold distribution. Most models stay close to the uniform 12.5% per letter, but several token-compression models over-predict early letters when uncertain, which is visible as a warm left edge and which the committee filtering in the main text does not remove because it is a model behavior rather than a data artifact. We release the full letter histograms so that any position-driven inflation can be audited per model. Figure 14b shows the pairwise correctness correlation between models, which averages $\phi = 0.24$ and reaches 0.54, so models tend to fail on the same questions rather than independently, and the two compression models with the strongest letter bias sit apart from the main cluster. Averaging every per-slice number over 22 models trades per-model resolution for a stable field-level estimate, so a slice describes the difficulty the benchmark poses to the field rather than any one system. Slices with fewer than 30 questions carry wide bootstrap intervals and are read as directional only.

E Identity conditioning pipelines

Each pipeline runs three stages before the VLM sees anything. A detector and tracker find and follow every person and animal, a cross-shot linker merges tracklets that belong to the same identity, and a grounding step maps the question referent to one global identity and renders a clip that marks only that identity. We cross four trackers, namely BoT-SORT (Aharon et al., 2022), ByteTrack (Zhang et al., 2022), DeepOCSORT (Maggiolino et al., 2023), and StrongSORT (Du et al., 2023), with five linkers, namely generic CLIP features (Radford et al., 2021) and the four dedicated re-identification models OSNet (Zhou et al., 2019), CLIP-ReID (Li et al., 2023b), TransReID (He et al., 2021), and SOLIDER (Chen et al., 2023), for twenty pipelines in total. CLIP linking is the generic baseline linker. It embeds the cropped frames of each tracklet with a frozen CLIP image encoder (Radford et al., 2021), averages them into one tracklet descriptor, and merges two tracklets when their descriptors exceed a cosine-similarity threshold, so it associates identities by general visual appearance rather than by a representation trained for re-identification. The four dedicated models replace this generic embedding with a re-identification embedding trained to be invariant to pose, viewpoint, and background, and are otherwise dropped into the

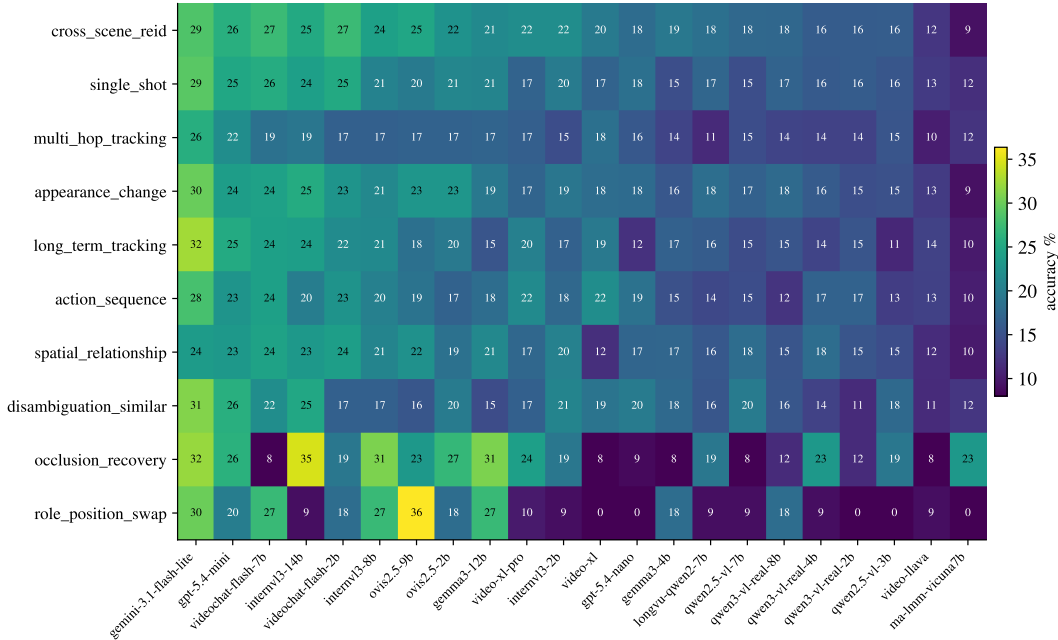


Figure 12: Accuracy of each model on each challenge type. Temporal tracking skills, namely role swaps, multi-hop tracking, and long-term tracking, are hardest across every model, and single-scene appearance types are easiest. This is the full per-model heatmap for the hierarchy summarized in Section 4.1.

same link-by-threshold step. Every pipeline is evaluated with all seventeen local models on a common question set, and Table 3 reports the mean over those models.

The highlighted raw-video row at 19.67% sits above every pipeline, so handing the model a clean identity track hurts rather than helps. The spread across trackers never exceeds 0.6 points, which shows the result is not an artifact of one weak tracker, and the dedicated re-identification models are marginally worse than generic CLIP linking, which shows stronger identity embeddings do not translate into better answers. We read this as evidence that capable VLMs exploit scene context that a single-identity crop removes, and that identity association is not the operation the benchmark is bottlenecked on. The grounding step is oracle-free, since the referent phrase is matched to a global identity by the same features that link tracklets, so no pipeline ever sees the gold answer. Rendering marks only the grounded identity by suppressing other regions, which is precisely the intervention that strips the scene context capable models rely on.

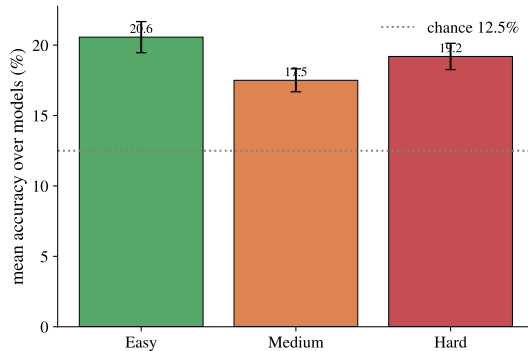
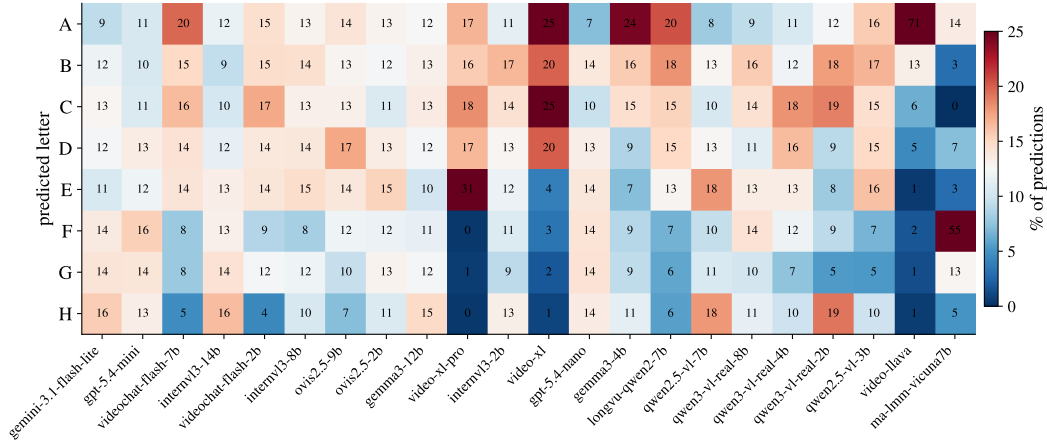
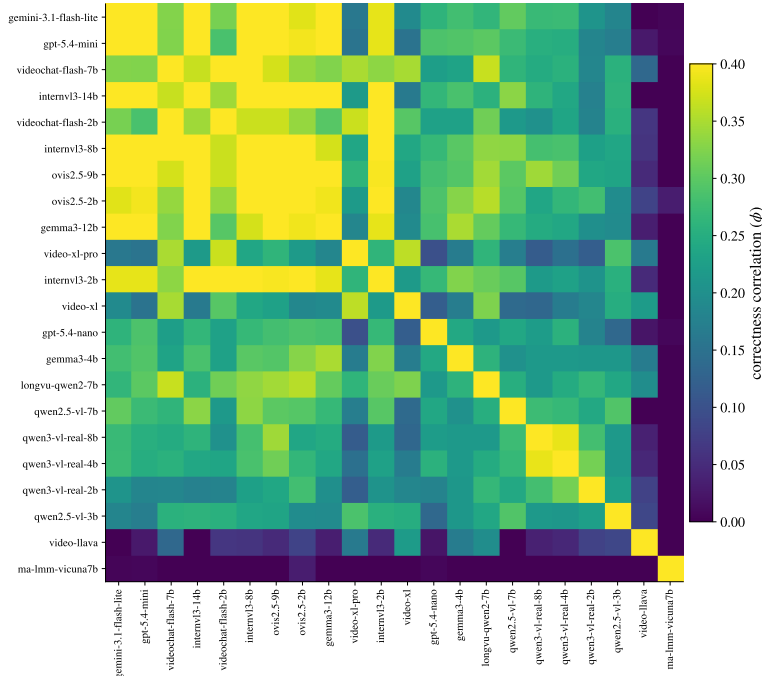


Figure 13: Mean accuracy over models by assigned difficulty against a 12.5% chance level. The relation to the assigned label is weak and non-monotonic, which motivates using measured consensus difficulty instead.



(a) Position bias



(b) Inter-model agreement

Figure 14: Top, predicted-letter distribution per model, where a uniform column is 12.5% per letter. Bottom, pairwise correctness correlation between models, mean $\phi = 0.24$.

F Selection probe and prior-work reconciliation

The selection probe scores 64 candidate frames per video against the question text with a CLIP encoder and keeps the eight highest-scoring frames, without any training. Figure 16 shows the per-model lift over uniform sampling at a fixed eight-frame budget, split into a question-conditioned variant that scores against the full question and a referent-conditioned variant that scores against the referent phrase only. Both variants raise accuracy on full-fidelity models by 1.0 to 2.7 points, which confirms that the right eight frames beat eight uniform frames. The one model that loses accuracy is VideoChat-Flash, which performs its own internal frame selection, so an external selector conflicts with its native handling and selection must instead be integrated inside the model.

Table 4 reconciles our numbers with published claims on the same source videos. A model family that reports 87.5% on original MovieChat-1k questions and a token-compression model that exceeds 95% on needle-in-a-haystack retrieval both fall below 20% on PersistQA,

Table 3: Accuracy averaged over seventeen VLMs on a common subset, for five identity-linking methods crossed with four trackers. The raw-video baseline is highlighted, and every pipeline sits at or below it.

Generic feature linking					
Method	BoT-SORT	ByteTrack	DeepOCSORT	StrongSORT	Mean
CLIP linking	19.00	18.94	18.70	18.90	18.89
Dedicated re-identification models					
OSNet	18.60	18.43	18.70	18.56	18.57
CLIP-ReID	18.82	18.47	18.23	18.65	18.54
TransReID	18.61	18.28	18.53	18.46	18.47
SOLIDER	18.73	18.45	18.28	18.51	18.49
Raw video	19.67				

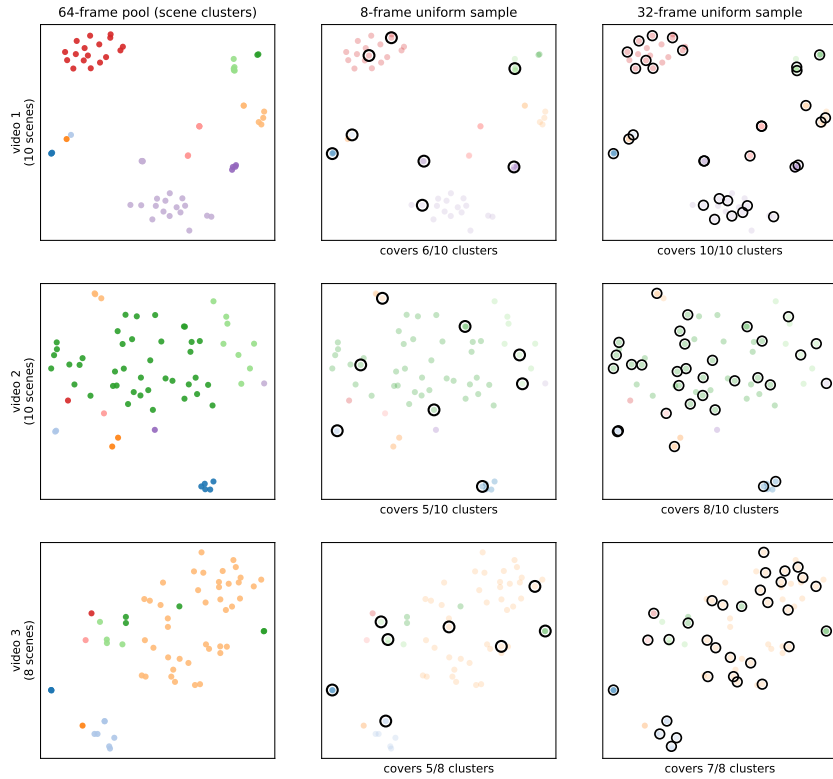


Figure 15: t-SNE of a 64-frame pool per video, where colors are scene clusters and markers show which clusters an 8-frame and a 32-frame uniform sample reach. The uncovered clusters visualize the coverage gap that Figure 6c quantifies.

which shows the gap is not explained by video difficulty or by the models being weak in general. The common factor is that prior tasks reward coarse retrieval and global gist, while PersistQA requires reading a fine-grained attribute of a tracked referent. Scale, which helps on many prior suites, is uncorrelated with accuracy here at $r = 0.05$, which is the last row of the table. The selector is training-free and adds no parameters, so its lift is a lower bound on what a learned in-model selector could reach. The prior-work rows quote published numbers on the same source videos rather than re-runs, so they establish a task gap rather than a controlled ablation.

Table 4: Published long video claims against PersistQA. The same source videos and models drop sharply once questions require identity-grounded reading.

Setting	Published	PersistQA
MovieChat-1k original QA (HierarQ)	87.5	<20
Needle-in-a-haystack (Video-XL-Pro)	>95	18–19
Long video suites, strong systems	>80	28.3 best
Scale helps accuracy	yes	$r = 0.05$

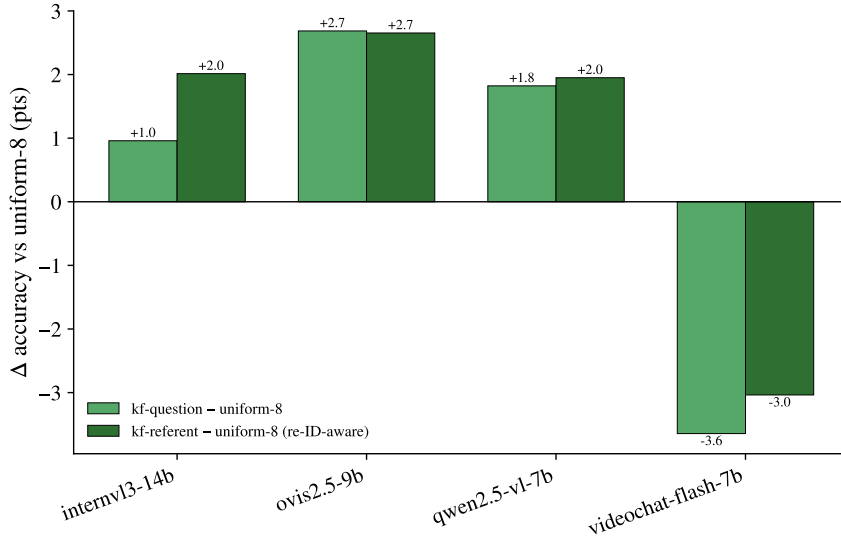


Figure 16: Lift of CLIP keyframe selection over uniform sampling per model. External selection hurts the model with native internal frame handling.

G Difficulty drivers across video properties

The main text rejects identity confusion as the source of difficulty using two proxies, so this appendix widens the test to every measurable video property. If difficulty were driven by how visually complex a video is, per-video accuracy would fall as duration, cuts, and variability rise, and if it were driven by identity confusion it would fall as concurrent people and identity-switch opportunity rise. We compute per-video accuracy averaged over all 22 models and correlate it against five properties, with quintile means overlaid in Figure 17. Difficulty rises slightly with duration at $r = +0.43$, scene cuts at $r = +0.31$, and visual variability at $r = +0.42$, and it is uncorrelated with the two identity-confusion proxies, at $r = +0.02$ for maximum concurrent people and $r = +0.03$ for identity-switch opportunity. The positive signs mean longer and more varied videos are marginally easier, not harder, and the near-zero identity proxies mean crowding a scene with people does not raise difficulty. This reinforces the main-text conclusion that the bottleneck is temporal coverage rather than identity confusion. Identity-switch opportunity counts frames where two tracked identities cross or exchange screen position, so it is a direct proxy for the confusion the benchmark name suggests, and its near-zero correlation is the cleanest single rejection of the identity-confusion reading. The positive duration and cut correlations are weak enough that they set no usable difficulty prior.

H Question-side difficulty

Difficulty could live in the question text rather than the video, so we test whether wording, referral strategy, or referent type predicts accuracy. If long or particular phrasings were the obstacle, accuracy would track question length or a single referral strategy. We measure accuracy against three question-side factors in Figure 18. Accuracy is flat against question length at Pearson $r = +0.02$, so wordier questions are not harder. Accuracy across the five well-supported referral strategies spans a narrow 17.6 to 19.4 band, with activity at 19.4, mixed at 18.4, environment at 18.2, location at 18.1, and appearance at 17.6, so how the

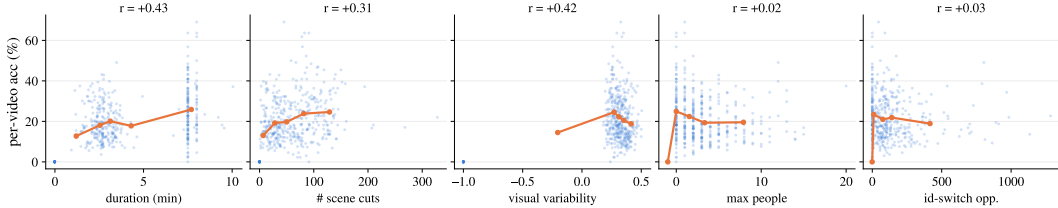


Figure 17: Per-video accuracy over 22 models against five video properties, with quintile means overlaid. Duration, scene cuts, and visual variability carry weak positive correlations, while the identity-confusion proxies of concurrent people and identity-switch opportunity are uncorrelated with difficulty.

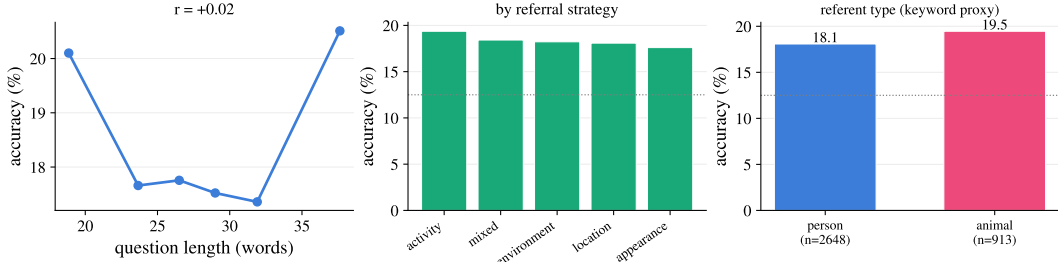


Figure 18: Question-side difficulty. Left, accuracy against question length is flat at $r = +0.02$. Middle, accuracy by referral strategy spans a narrow 17.6 to 19.4 band. Right, a keyword proxy for referent type gives person 18.1% and animal 19.5%.

referent is named barely changes difficulty. Splitting by referent type through a keyword proxy, since no referent-type metadata tag exists, gives person referents 18.1% at $n = 2648$ against animal referents 19.5% at $n = 913$, so animal referents are marginally easier. No question-side factor concentrates the difficulty, which points back to the visual tracking demand as the shared cause. The referent-type split uses a keyword proxy rather than a metadata tag, so its person and animal counts are approximate and the 1.4-point gap is read as directional. No question-side factor exceeds a weak correlation, which rules out phrasing as a hidden text-only shortcut the committee missed.

I Coverage across models

The main text shows coverage helps InternVL3-14B, so this appendix checks that the coverage lever generalizes across the model set rather than being one family’s quirk. We sweep the uniform frame budget from 8 to 64 for seven full-fidelity models in Figure 19. Coverage helps most models, with InternVL3-14B gaining 3.2 points and Qwen2.5-VL-7B gaining 3.1 points from 8 to 32 frames, and one model, Qwen3-VL-8B, stays flat across the sweep. InternVL3-8B and InternVL3-14B have no valid 64-frame point because that run broke, so their curves stop at 32 frames. The consistent upward slope across families confirms that added temporal coverage, not a single architecture, is the shared lever, while the one flat model marks the ceiling of what uniform coverage alone can buy. The sweep holds every other setting fixed, so the slope isolates temporal coverage from resolution and prompt. The flat model marks where uniform coverage alone stops paying, which is the exact gap the query-conditioned selection of Section 5 targets.

J Scaling within families

The main text reports a weak parameter-to-accuracy correlation of 0.05 across all models, which could hide a strong within-family trend masked by family differences. We plot accuracy against parameter count with a separate line per family in Figure 20. Scaling inside a family adds a few points at most, and the vertical gaps between family lines exceed the slope along any one of them, so family identity dominates parameter count. This within-family view echoes the main-text $r = 0.05$ and shows that the weak overall correlation is not an averaging artifact but a real ceiling on what scale delivers for this skill. Reading slope

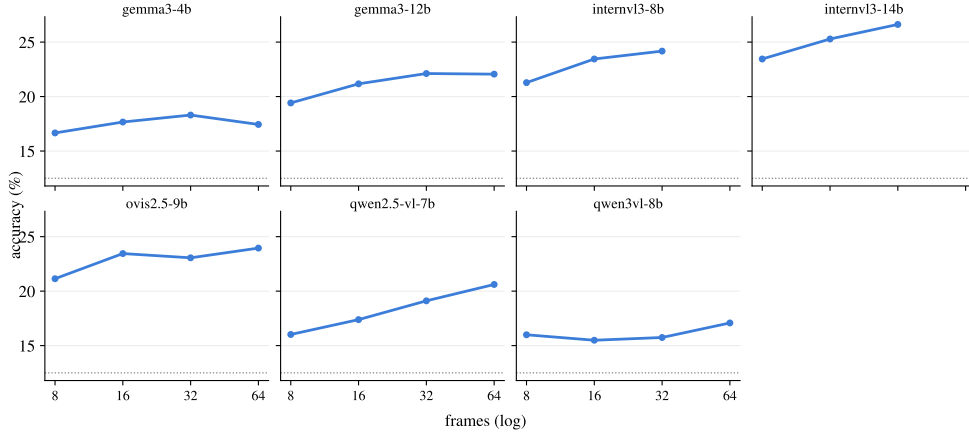


Figure 19: Accuracy against uniform frame budget for seven full-fidelity models, one panel each. Most models rise with more frames, InternVL3-14B by 3.2 points and Qwen2.5-VL-7B by 3.1 points from 8 to 32 frames, while Qwen3-VL-8B is flat. The InternVL3-8B and 14B curves stop at 32 frames because the 64-frame run broke.

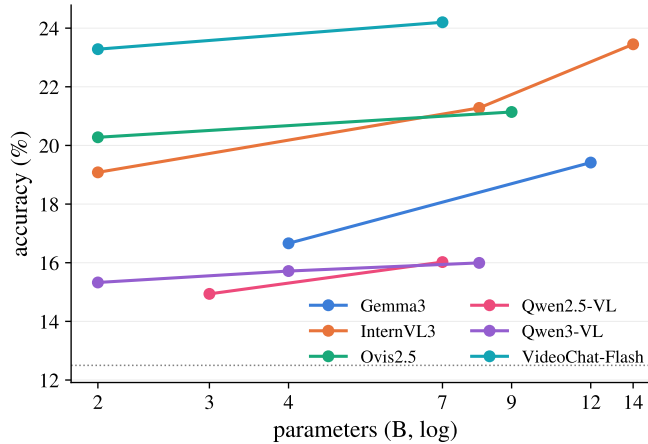


Figure 20: Accuracy against parameter count with a within-family line per model family. Scaling inside a family adds a few points at most, and family identity dominates parameter count, echoing the main-text correlation of 0.05.

within a family removes the confound that families differ in data and training recipe, which an all-model regression would fold into the trend. The residual within-family slope stays under a few points, so parameter count is not a hidden lever masked by family mixing.

K Per-challenge coverage gain

Coverage helps unevenly across challenge types in Section 4.5, so this appendix quantifies which challenge types gain the most from added frames. We compute the mean accuracy gain from 1 to 8 frames for each challenge type, averaged over the four frame-control models, and render it as a one-row heatmap in Figure 21. The types that require re-locating an identity after it leaves view, namely occlusion recovery and disambiguation among similar identities, gain the most from added frames, while types answerable from a single reference frame gain little. This localizes the coverage benefit to exactly the tracking-heavy skills the benchmark targets, so the coverage lever acts where the benchmark is hardest. The gain is measured from 1 to 8 frames, so it captures the move from a single reference frame to a short temporal window rather than the full sweep. Occlusion recovery and disambiguation gain most because their answer frame is often absent from a single-frame sample, which is the coverage failure the main text names.

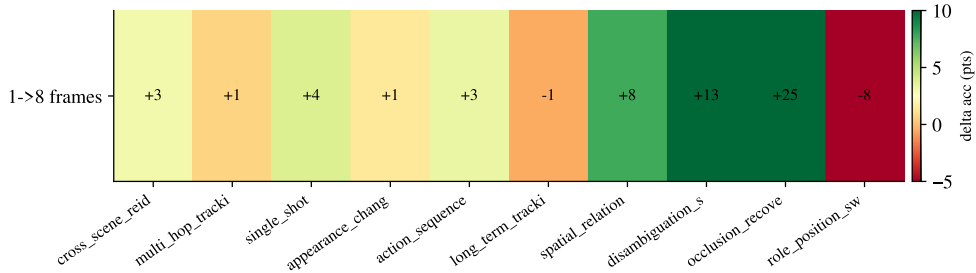


Figure 21: Mean accuracy gain from 1 to 8 frames per challenge type, averaged over the four frame-control models. Types that require re-locating a lost identity gain the most from added frames.

L Robustness batteries

Section 4.5 summarizes three robustness probes, and this appendix shows each in full on the four frame-control models. The probes matter because a benchmark whose answers survive frame shuffling or option rotation would be measuring a shortcut rather than temporal reading, so we test frame order, option order, and input corruption directly.

Frame order carries a weak signal that shrinks with scale. Figure 8a shows that shuffling or reversing the eight frames lowers InternVL3-2B by 1.8 to 2.8 points, while the effect reaches about zero for InternVL3-14B. Option order barely moves accuracy yet flips the chosen answer often. Figure 8b shows that rotating the option contents by three positions leaves accuracy near flat while the model changes its content answer on 49.8% of questions, so accuracy stability hides an unstable choice. Both results are consistent with models reading frames as a loosely ordered set rather than a tracked sequence, which matches the tracking failures in Section 4.1. Input corruption lowers accuracy while losing pixels does not. If the benchmark were solvable from coarse global cues, mild corruption would leave accuracy untouched, so we probe six corruptions against a clean baseline. We apply blur, JPEG, Gaussian noise, low light, motion blur, and downscaling to 112 pixels at a fixed reduced-resolution probe setting, shown in the main-text Figure 8(c). Because the clean baseline itself runs at reduced resolution, the panels report relative drops rather than absolute accuracy. Corruption lowers accuracy by up to 4.5 points, moving InternVL3-2B from 17.8% clean to 13.2% under noise, while downscaling to 112 pixels is nearly lossless. The pattern matches the weak per-frame resolution effect in Section 4.4, since losing pixels barely hurts while losing signal content does, which is consistent with coverage rather than fidelity being the bottleneck. Because these probes ablate one factor at a time against a shared clean baseline, they isolate whether each corruption removes benchmark-relevant signal, and only content-destroying corruptions do.

M Visual content landscape

A natural worry is that difficulty is tied to a particular kind of video, such as dark or crowded footage, which would make PersistQA a test of one visual regime rather than of identity-grounded reading. We embed every video by its mean CLIP frame feature, project the set with t-SNE, and color each point by per-video accuracy averaged over all 22 models in Figure 22. Accuracy is mixed across the visual-content clusters with no accuracy tiers, so easy and hard videos sit side by side within every cluster and difficulty is not tied to a particular visual video type. This contrasts honestly with robustness studies that report clean accuracy tiers by content, and it means PersistQA difficulty is a property of the identity-grounded question rather than of the footage it is asked over. Coloring by per-video accuracy rather than by content lets any cluster reveal an accuracy tier it carries, and none appear. This is a stronger control than reporting mean accuracy per content bin, since it would expose even a small pocket of uniformly hard footage.

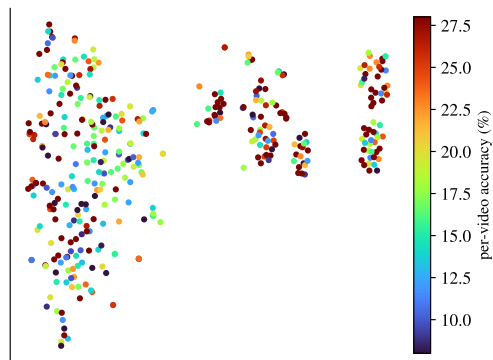


Figure 22: t-SNE of every video by mean CLIP frame embedding, colored by per-video accuracy over 22 models. Accuracy is mixed within every visual-content cluster with no accuracy tiers, so difficulty is not tied to a particular visual video type.